

Part II

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- a) What is the risk of melt-down in the power plant during a day if no observations have been made? What if there is icy weather?

$$P(\text{Meltdown}) = 0.02578$$

$$P(\text{Meltdown} | \text{IcyWeather}) = 0.03472$$

- b) Suppose that both warning sensors indicate failure. What is the risk of a meltdown in that case? Compare this result with the risk of a melt-down when there is an actual pump failure and water leak. What is the difference? The answers must be expressed as conditional probabilities of the observed variables, $P(\text{Meltdown} | \dots)$.

$$P(\text{Meltdown} | \text{PumpFailureWarning}, \text{WaterLeakWarning}) = 0.14535$$

$$P(\text{Meltdown} | \text{PumpFailure}, \text{WaterLeak}) = 0.2$$

The difference is that the first probability is conditioned on noisy sensor observations, while the second is conditioned on true mechanical failures.

- c) The conditional probabilities for the stochastic variables are often estimated by repeated experiments or observations. Why is it sometimes very difficult to get accurate numbers for these? What conditional probabilities in the model of the plant do you think are difficult or impossible to estimate?

It is difficult to obtain accurate conditional probabilities because some events, such as mechanical failures or meltdowns, are rare or dangerous, making repeated experiments impossible. In the plant model, probabilities related to major failures or meltdowns are the hardest to estimate.

- d) Assume that the "IcyWeather" variable is changed to a more accurate "Temperature" variable instead (don't change your model). What are the different alternatives for the domain of this variable? What will happen with the probability distribution of $P(\text{WaterLeak} | \text{Temperature})$ in each alternative?

The domain of this variable can use different temperature scales (e.g. Fahrenheit, Celsius or Kelvin). The probability of $P(\text{WaterLeak} | \text{Temperature})$ will be the same in different temperature scales, if the temperature values correspond to each other in those scales.

$$\text{Example: } P(\text{Temperature} = 0\text{Celsius}) = P(\text{Temperature} = 32\text{Fahrenheit}) = P(\text{Temperature} = 273.15\text{Kelvin})$$

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- a) What does a probability table in a Bayesian network represent?

A probability table in a Bayesian network represents the conditional probabilities of a variable given all possible combinations of its parent variables. For a root node (with no parents), the table simply defines its prior probability distribution.

- b) What is a joint probability distribution? Using the chain rule on the structure of the Bayesian network to rewrite the joint distribution as a product of $P(\text{child}|\text{parent})$ expressions, calculate manually the particular entry in the joint distribution of $P(\text{Meltdown}=F, \text{PumpFailureWarning}=F, \text{PumpFailure}=F, \text{WaterLeakWarning}=F, \text{WaterLeak}=F, \text{IcyWeather}=F)$. Is this a common state for the nuclear plant to be in?

In a Bayesian network, the joint distribution can be written as the product of conditional probabilities of all nodes.

$$\begin{aligned}
 &P(\text{IcyWeather}, \text{WaterLeak}, \text{PumpFailure}, \text{WaterLeakWarning}, \text{PumpingFailureWarning}, \text{Meltdown}) \\
 &= P(\text{IcyWeather}) \cdot P(\text{WaterLeak} \mid \text{IcyWeather}) \cdot P(\text{PumpFailure}) \cdot P(\text{WaterLeakWarning} \mid \\
 &\text{WaterLeak}) \cdot P(\text{PumpFailureWarning} \mid \text{PumpFailure}) \cdot P(\text{Meltdown} \mid \text{PumpFailure}, \text{WaterLeak}) \\
 &= P(\text{IcyWeather} = F) \cdot P(\text{WaterLeak} = F \mid \text{IcyWeather} = F) \cdot P(\text{PumpFailure} = F) \cdot P(\text{WaterLeakWarning} = F \mid \\
 &\text{WaterLeak} = F) \cdot P(\text{PumpFailureWarning} = F \mid \text{PumpFailure} = F) \cdot P(\text{Meltdown} = F \mid \text{PumpFailure} = F, \\
 &\text{WaterLeak} = F) \\
 &= 0.95 \cdot 0.9 \cdot 0.9 \cdot 0.95 \cdot 0.95 \cdot 0.999 \\
 &\approx 0.69
 \end{aligned}$$

Yes, it is indeed a common and expected state for the nuclear plant to be in.

- c) What is the probability of a meltdown if you know that there is both a water leak and a pump failure? Would knowing the state of any other variable matter? Explain your reasoning?

$$P(\text{Meltdown} \mid \text{PumpFailure}, \text{WaterLeak}) = 0.2$$

No, because *Meltdown* depends directly only on *PumpFailure* and *WaterLeak* in the network structure.

- d) Calculate manually the probability of a meltdown when you happen to know that *PumpFailureWarning*=F, *WaterLeak*=F, *WaterLeakWarning*=F and *IcyWeather*=F but you are not really sure about a pump failure.

We need to compute

$$P(\text{Meltdown} = T \mid \text{PumpFailureWarning} = F, \text{WaterLeak} = F, \text{WaterLeakWarning} = F, \text{IcyWeather} = F)$$

Where *PumpFailure* is unobserved.

Inference by Enumeration

$$P(X \mid e) = \alpha \cdot P(X, e) = \alpha \cdot \sum_y P(X, e, y)$$

$$X = \{Meltdown\}$$

$$E = \{PumpFailureWarning, WaterLeak, WaterLeakWarning, IcyWeather\}$$

$$e = \{PumpFailureWarning = F, WaterLeak = F, WaterLeakWarning = F, IcyWeather = F\}$$

$$Y = \{PumpFailure\}$$

$$P(X | e) = \alpha \cdot \{ P(Meltdown, WaterLeak = F, PumpFailureWarning = F, WaterLeakWarning = F, IcyWeather = F, PumpFailure = F) + P(Meltdown, WaterLeak = F, PumpFailureWarning = F, WaterLeakWarning = F, IcyWeather = F, PumpFailure) \}$$

$$= \alpha \cdot \{ (P(Meltdown | PumpFailureWarning = F, WaterLeakWarning = F, PumpFailure = F, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailureWarning = F | WaterLeakWarning = F, PumpFailure = F, WaterLeak = F, IcyWeather = F) \cdot P(WaterLeakWarning = F | PumpFailure = F, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailure = F | WaterLeak = F, IcyWeather = F) \cdot P(WaterLeak = F | IcyWeather = F) \cdot P(IcyWeather = F) + (P(Meltdown | PumpFailureWarning = F, WaterLeakWarning = F, PumpFailure, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailureWarning = F | WaterLeakWarning = F, PumpFailure, WaterLeak = F, IcyWeather = F) \cdot P(WaterLeakWarning = F | PumpFailure, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailure | WaterLeak = F, IcyWeather = F) \cdot P(WaterLeak = F | IcyWeather = F) \cdot P(IcyWeather = F)) \}$$

$$= \alpha \cdot 0.00191$$

$$P(X = F | e) = \alpha \cdot \{ P(Meltdown = F, WaterLeak = F, PumpFailureWarning = F, WaterLeakWarning = F, IcyWeather = F, PumpFailure = F) + P(Meltdown = F, WaterLeak = F, PumpFailureWarning = F, WaterLeakWarning = F, IcyWeather = F, PumpFailure) \}$$

$$= \alpha \cdot \{ (P(Meltdown = F | PumpFailureWarning = F, WaterLeakWarning = F, PumpFailure = F, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailureWarning = F | WaterLeakWarning = F, PumpFailure = F, WaterLeak = F, IcyWeather = F) \cdot P(WaterLeakWarning = F | PumpFailure = F, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailure = F | WaterLeak = F, IcyWeather = F) \cdot P(WaterLeak = F | IcyWeather = F) \cdot P(IcyWeather = F) + (P(Meltdown = F | PumpFailureWarning = F, WaterLeakWarning = F, PumpFailure, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailureWarning = F | WaterLeakWarning = F, PumpFailure, WaterLeak = F, IcyWeather = F) \cdot P(WaterLeakWarning = F | PumpFailure, WaterLeak = F, IcyWeather = F) \cdot P(PumpFailure | WaterLeak = F, IcyWeather = F) \cdot P(WaterLeak = F | IcyWeather = F) \cdot P(IcyWeather = F)) \}$$

$$= \alpha \cdot 0.70068$$

$$\alpha = \frac{1}{P(X|e) + P(X = F|e)} = \frac{1}{0.00191 + 0.70068} = \frac{1}{0.70259} \approx 1.42331$$

$$P(X|e) = \alpha \cdot 0.00191 = 1.42331 \cdot 0.00191 \approx 0.0027$$

Part III

2

- a) During the lunch break, the owner tries to show off for his employees by demonstrating the many features of his car stereo. To everyone's disappointment, it doesn't work. How did the owner's chances of surviving the day change after this observation?

The owner's chances of surviving changed from $P(\text{Survives} = T) = 0.9904$ to $P(\text{Survives} = T \mid \text{Radio} = F) = 0.98304$.

- b) The owner buys a new bicycle that he brings to work every day. The bicycle has the following properties:

$$P(\text{bicycle_works}) = 0.9$$

$$P(\text{survives} \mid \neg \text{moves} \wedge \text{melt-down} \wedge \text{bicycle_works}) = 0.6$$

$$P(\text{survives} \mid \text{moves} \wedge \text{melt-down} \wedge \text{bicycle_works}) = 0.9$$

How does the bicycle change the owner's chances of survival?

The owner's chances of survival increased with $P(\text{Survives} = T) = 0.99522 - P(\text{Survives} = T) = 0.9904 = 0.00482$

- c) It is possible to model any function in propositional logic with Bayesian Networks. What does this fact say about the complexity of exact inference in Bayesian Networks? What alternatives are there to exact inference?

Yes, this means that exact inference in Bayesian Network is very complex, the time and space complexity increases with the complexity of the structure of the Bayesian Network.

As alternatives, we can use approximate inference methods, such as variational inference to get faster but less exact results.

Part IV

2

- a) The owner had an idea that instead of employing a safety person, to replace the pump with a better one. Is it possible, in your model, to compensate for the lack of Mr H.S.'s expertise with a better pump?
Yes, if there is a better pump, it means less failures, then there would be less warnings, which can compensate for the lack of Mr. H.S.'s expertise by reducing his action.
- b) Mr H.S. fell asleep on one of the plant's couches. When he wakes up he hears someone scream: "There is one or more warning signals beeping in your control room!". Mr H.S. realizes that he does not have time to fix the error

before it is too late (we can assume that he wasn't in the control room at all). What is the chance of survival for Mr H.S. if he has a car with the same properties as the owner?

Given the evidence $isWarning = T$ and $Sleeping = T$, the network yields $P(Survives) = 0.98081$

- c) What unrealistic assumptions do you make when creating a Bayesian Network model of a person?

We assume that a person will consistently act in a certain way with a certain probability with no sudden unforeseen medical or mental changes. For example, Mr. H.S. may suddenly have a panic attack and cannot handle the alarms, even if his mental status is normal and he never had this problem before.

- d) Describe how you would model a more dynamic world where for example the "IcyWeather" is more likely to be true the next day if it was true the day before. You only have to consider a limited sequence of days.

We can add a node which contains information about icy weather in previous days. This node can affect the node *IcyWeather*, which represents today's chance of icy weather.